

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409940
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Durocher; Eric S. et al.

Electric motor propulsion system for an aircraft

Abstract

An assembly for a propulsion system of an aircraft includes a gearbox module, an electric motor assembly, a first accessory load assembly, and a propulsor. The gearbox module includes a gear assembly and an output shaft. The gear assembly includes a main gear and a plurality of offset gears. The electric motor assembly includes an electric motor. The electric motor includes a rotor. The rotor is mounted to a first offset gear of the plurality of offset gears. The rotor is configured for rotation about a rotational axis to drive rotation of the gear assembly and the output shaft. The first accessory load assembly includes at least one first accessory load. The at least one first accessory load is mounted to a second offset gear of the plurality of offset gears. The propulsor is coupled to the output shaft.

Inventors: Durocher; Eric S. (Boucherville, CA), Labrecque; Michel (Ste-Julie, CA)

Applicant: Pratt & Whitney Canada Corp. (Longueuil, CA)

Family ID: 91580783

Assignee: Pratt & Whitney Canada Corp. (Longueuil, CA)

Appl. No.: 18/209892

Filed: June 14, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240417083 A1	Dec. 19, 2024

Publication Classification

Int. Cl.: B64D27/24 (20240101); B64D35/00 (20060101); F16H1/22 (20060101); F16H57/038 (20120101); F16H57/04 (20100101); H02K7/116 (20060101); F16H57/02 (20120101)

U.S. Cl.:

CPC **B64D27/24** (20130101); **B64D35/00** (20130101); **F16H1/22** (20130101); **F16H57/038** (20130101); **F16H57/0441** (20130101); **H02K7/116** (20130101); F16H2057/02026 (20130101); F16H2057/02034 (20130101); F16H2057/02043 (20130101); F16H2057/02082 (20130101)

Field of Classification Search

CPC: B64D (35/021); B64D (35/00); B64D (27/02); B64D (27/24); B64D (27/30); F02C (7/32)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5782433	12/1997	Goi	475/216	B64C 27/14
8324746	12/2011	Bradbrook	290/1A	F02C 3/10
10273883	12/2018	Roach	N/A	N/A
11415061	12/2021	De Wergifosse	N/A	B60K 6/36
11485503	12/2021	Dubreuil	N/A	N/A
2020/0307812	12/2019	Dubreuil	N/A	B64D 35/08
2021/0179286	12/2020	Harvey	N/A	N/A
2021/0276706	12/2020	Heironimus	N/A	B64C 27/68
2021/0388732	12/2020	Turcotte	N/A	F01D 15/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3146966	12/2023	FR	N/A

OTHER PUBLICATIONS

EP Search Report for EP Patent Application No. 24182372.3 dated Oct. 25, 2024. cited by applicant

Primary Examiner: Abell; Tye William

Attorney, Agent or Firm: Getz Balich LLC

Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to an aircraft propulsion system and, more particularly, to an electric aircraft propulsion system driven by an electric motor.

BACKGROUND OF THE ART

(2) Electric propulsion systems for aircraft may include an electric motor configured to provide thrust for the aircraft. Various electric aircraft propulsion systems are known in the art. While these known systems have various advantages, there is still room in the art for improvement.

SUMMARY

(3) It should be understood that any or all of the features or embodiments described herein can be

used or combined in any combination with each and every other feature or embodiment described herein unless expressly noted otherwise.

(4) According to an aspect of the present disclosure, an assembly for a propulsion system of an aircraft includes a gearbox module, an electric motor assembly, a first accessory load assembly, and a propulsor. The gearbox module includes a gear assembly and an output shaft. The gear assembly includes a main gear and a plurality of offset gears. The plurality of offset gears are engaged with the main gear. The output shaft and the main gear are configured for rotation about an axial centerline of the gear assembly. The electric motor assembly includes an electric motor. The electric motor includes a rotor. The rotor is mounted to a first offset gear of the plurality of offset gears. The rotor is configured for rotation about a rotational axis to drive rotation of the gear assembly and the output shaft. The first accessory load assembly includes at least one first accessory load. The at least one first accessory load is mounted to a second offset gear of the plurality of offset gears. The propulsor is coupled to the output shaft.

(5) In any of the aspects or embodiments described above and herein, the rotational axis may be radially offset from the axial centerline.

(6) In any of the aspects or embodiments described above and herein, the output shaft may be fixedly mounted to the main gear.

(7) In any of the aspects or embodiments described above and herein, the first accessory load assembly may be axially coincident with the electric motor assembly.

(8) In any of the aspects or embodiments described above and herein, the gearbox module may include a gearbox housing surrounding the gear assembly and the output shaft, the electric motor assembly may include a motor housing, and the motor housing may be mounted to the gearbox housing.

(9) In any of the aspects or embodiments described above and herein, the electric motor assembly may further include a mount assembly mounting the motor housing to the gearbox housing. The mount assembly may include a motor bracket and a plurality of struts. The motor bracket may be mounted to the motor housing at an aft end of the motor housing. Each of the plurality of struts may extend between and connect the motor bracket to the gearbox housing.

(10) In any of the aspects or embodiments described above and herein, the assembly may further include a second accessory load assembly including at least one second accessory load. The at least one second accessory load may be engaged with the output shaft.

(11) In any of the aspects or embodiments described above and herein, the second accessory load assembly may be axially coincident with the electric motor assembly.

(12) In any of the aspects or embodiments described above and herein, the first accessory load assembly may include an oil pump disposed at a vertical bottom of the gearbox module.

(13) In any of the aspects or embodiments described above and herein, the gearbox module may include a gearbox housing surrounding the gear assembly and the output shaft. The oil pump may be disposed within the gearbox housing.

(14) In any of the aspects or embodiments described above and herein, the output shaft may include a bevel gear. The assembly may further include a second accessory load assembly including at least one second accessory load. The at least one second accessory load may be engaged with the bevel gear.

(15) In any of the aspects or embodiments described above and herein, the gear assembly may further include a ring gear engaged with each of the offset gears. The ring gear may be mounted to the output shaft and configured to drive the output shaft for rotation about the axial centerline.

(16) According to another aspect of the present disclosure, an assembly for a propulsion system of an aircraft includes a propulsor, a gearbox module, an electric motor assembly, and a plurality of accessory load assemblies. The gearbox module includes a gear assembly and an output shaft. The gear assembly includes a main gear and a plurality of offset gears. The main gear is configured for rotation about an axial centerline of the gear assembly. The plurality of offset gears are engaged

with the main gear. The plurality of offset gears includes a first offset gear and a plurality of second offset gears. The output shaft extends along the axial centerline and interconnects the propulsor and the main gear. The electric motor assembly includes an electric motor. The electric motor includes a rotor. The rotor is mounted to the first offset gear. The rotor is configured for rotation about a rotational axis to drive rotation of the first offset gear. Each accessory load assembly of the plurality of accessory load assemblies includes at least one accessory load coupled to a respective second offset gear of the plurality of second offset gears.

(17) In any of the aspects or embodiments described above and herein, the plurality of accessory load assemblies may be axially coincident with the electric motor assembly.

(18) In any of the aspects or embodiments described above and herein, each offset gear of the plurality of offset gears may be disposed radially adjacent the main gear with respect to the axial centerline.

(19) In any of the aspects or embodiments described above and herein, the rotational axis may be radially offset from the axial centerline.

(20) According to another aspect of the present disclosure, an assembly for a propulsion system of an aircraft includes a propulsor, a gearbox module, an electric motor assembly, and a plurality of accessory load assemblies. The gearbox module includes a gear assembly and an output shaft. The gear assembly includes a main gear and a plurality of offset gears. The plurality of offset gears are engaged with the main gear. The plurality of offset gears includes a first offset gear and a plurality of second offset gears. The output shaft interconnects the propulsor and the main gear. The electric motor assembly includes an electric motor. The electric motor includes a rotor. The rotor is mounted to the first offset gear. The rotor is configured for rotation about a rotational axis to drive rotation of the first offset gear. Each accessory load assembly of the plurality of accessory load assemblies includes at least one accessory load coupled to a respective gear of the main gear or the plurality of offset gears.

(21) In any of the aspects or embodiments described above and herein, the gearbox module may include a gearbox housing surrounding the gear assembly and the output shaft.

(22) In any of the aspects or embodiments described above and herein, the plurality of accessory load assemblies may include a first accessory load assembly including a first accessory load and a second accessory load. The first accessory load may be disposed within the gearbox housing and the second accessory load may be disposed outside the gearbox housing.

(23) In any of the aspects or embodiments described above and herein, the first accessory load may be an oil pump disposed at a vertical bottom of the gearbox housing.

(24) The present disclosure, and all its aspects, embodiments and advantages associated therewith will become more readily apparent in view of the detailed description provided below, including the accompanying drawings.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 schematically illustrates a side, cutaway view of a portion of an aircraft including an electric propulsion system, in accordance with one or more embodiments of the present disclosure.

(2) FIG. 2 schematically illustrates a side, cutaway view of a rotational assembly for an electric propulsion system, in accordance with one or more embodiments of the present disclosure.

(3) FIG. 3 schematically illustrates a front, cutaway view of the rotational assembly of FIG. 2, in accordance with one or more embodiments of the present disclosure.

(4) FIG. 4 schematically illustrates a side, cutaway view of another rotational assembly for an electric propulsion system including a mount assembly, in accordance with one or more embodiments of the present disclosure.

(5) FIG. 5 schematically illustrates a front, cutaway view of the rotational assembly of FIG. 4, in accordance with one or more embodiments of the present disclosure.

(6) FIG. 6 schematically illustrates a side, cutaway view of another rotational assembly for an electric propulsion system, in accordance with one or more embodiments of the present disclosure.

(7) FIG. 7 schematically illustrates a front, cutaway view of another rotational assembly for an electric propulsion system, in accordance with one or more embodiments of the present disclosure.

(8) FIG. 8 schematically illustrates a side, cutaway view of another rotational assembly for an electric propulsion system, in accordance with one or more embodiments of the present disclosure.

(9) FIG. 9 schematically illustrates a side, cutaway view of a portion of an aircraft including a gas turbine engine propulsion system, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(10) FIG. 1 schematically illustrates a propulsion system **20** configured for an aircraft, such as the aircraft **1000** of FIG. 1. The aircraft **1000** may be a fixed-wing aircraft (e.g., an airplane), as shown in FIG. 1. Moreover, the aircraft may be a manned aerial vehicle or an unmanned aerial vehicle (UAV, e.g., a drone).

(11) The propulsion system **20** of FIG. 1 includes a rotational assembly **22**, one or more accessory load assemblies **24**, and an electrical power source **26**. The rotational assembly **22** includes an electric motor assembly **28**, a transmission gearbox (TGB) module **32**, and a propulsor **34** (e.g., a propeller).

(12) The electric motor assembly **28** includes an electric motor **38** and a motor housing **40**. The electric motor **38** is electrically connected to the electrical power source **26**. For example, the electric motor **38** is configured to receive electrical power from the electrical power source **26** for operation of the electric motor **38**. The electric motor **38** may additionally be configured to direct electrical power to the electrical power source **26** (e.g., the electric motor **38** may be configured to operate as an electrical generator). The electric motor **38** may be configured as an alternating current (AC) electric motor or a direct current (DC) electric motor. For example, the electric motor **38** may be configured as a synchronous electric motor, an induction motor (e.g., an asynchronous electric motor), a permanent magnet electric motor, or the like, and the present disclosure is not limited to any particular electric motor configuration. The electric motor assembly **28** may further include other electrical control and/or conditioning equipment and components such as, but not limited to, an inverter, a motor controller, AC/DC conversion equipment, and/or the like.

(13) The electric motor **38** of FIG. 1 includes a rotor **42** and a stator **44**. The rotor **42** is configured for rotation about a rotational axis **36**. The rotor **42** includes a rotor body **46**. The rotor **42** may additionally include a shaft **48**. The rotor body **46** is disposed in the motor housing **40**. The rotor body **46** extends (e.g., axially extends) between and to a first axial end **52** of the rotor body **46** and a second axial end **54** of the rotor body **46**. The shaft **48** is disposed at (e.g., on, adjacent, or proximate) the first axial end **52**. The shaft **48** extends along the rotational axis **36** axially outward from the first axial end **52**. For example, the shaft **48** may extend through the motor housing **40** to the TGB module **32**. The stator **44** of FIG. 1 extends circumferentially about (e.g., completely around) the rotational axis **36** and the rotor body **46** in the motor housing **40**.

(14) The motor housing **40** is configured to support and house the electric motor **38**. The motor housing **40** may extend circumferentially about (e.g., completely around) the rotational axis **36**. The motor housing **40** extends (e.g., axially extends) between and to a first axial end **56** of the motor housing **40** and a second axial end **58** of the motor housing **40**. The motor housing **40** may be attached to the TGB module **32**, for example, at (e.g., on, adjacent, or proximate) the first axial end **56**. The motor housing **40** may form or otherwise include a flange or other suitable mounting configuration for mounting the motor housing **40** to the TGB module **32**. The motor housing **40** may additionally include one or more bearing assemblies for rotatably supporting the rotor **42** and its shaft **48**.

(15) FIGS. 2 and 3 schematically illustrate cutaway views of portions of the TGB module 32 coupled to accessory load assemblies 24 and the electric motor assembly 28. The TGB module 32 includes a gearbox housing 60, an output shaft 62, and a gear assembly 64.

(16) The gearbox housing 60 may be mounted to or otherwise disposed at (e.g., on, adjacent, or proximate) the electric motor assembly 28 (e.g., the motor housing 40 at the first axial end 56; see FIG. 1). The gearbox housing 60 may extend about (e.g., completely around) an axial centerline 66 of the gear assembly 64. The gearbox housing 60 may surround and provide support for the output shaft 62 and the gear assembly 64. The gearbox housing 60 may additionally provide support for the accessory load assemblies 24, as will be discussed in further detail. The gearbox housing 60 may additionally include or otherwise support one or more bearing assemblies for rotatably supporting the output shaft 62. The gearbox housing 60 may form an oil cavity 61 (e.g., an oil sump) within the gearbox housing 60. The oil cavity 61 may be formed by the gearbox housing 60 vertically below the output shaft 62, for example, at a vertical bottom of the gearbox housing 60 (e.g., with respect to a normal attitude of the propulsion system 20; see FIG. 1). Oil from the electric motor 38, the output shaft 62, the gear assembly 64, bearing assemblies, and/or other oil-lubricated and/or oil-cooled components of the rotational assembly 22 may drain to or otherwise be directed to the oil cavity 61.

(17) The output shaft 62 extends circumferentially about (e.g., completely around) the axial centerline 66. The axial centerline 66 of FIGS. 2 and 3 is a rotational axis of the output shaft 62. The output shaft 62 extends between and to a first axial end 68 of the output shaft 62 and a second axial end 70 of the output shaft 62. The first axial end 68 is mounted to or otherwise disposed at (e.g., on, adjacent, or proximate) the propulsor 34 such that rotation of the output shaft 62 about the axial centerline 66 drives rotation of the propulsor 34.

(18) The gear assembly 64 of FIGS. 2 and 3 includes a main gear 72 and a plurality of offset gears 74. The main gear 72 extends circumferentially about (e.g., completely around) the axial centerline 66. The axial centerline 66 of FIGS. 2 and 3 is a rotational axis of the main gear 72. The main gear 72 is fixedly mounted to the output shaft 62. For example, the main gear 72 may be mounted to the output shaft 62 at (e.g., on, adjacent, or proximate) the second axial end 70. Alternatively, the main gear 72 may be mounted to an axially intermediate portion of the output shaft 62 between the first axial end 68 and the second axial end 70. The main gear 72 circumscribes the output shaft 62. The main gear 72 is configured for rotation about the axial centerline 66 with the output shaft 62. The main gear 72 includes gear teeth 76 engaged with (e.g., meshed with) each of the offset gears 74. The offset gears 74 are radially offset from the axial centerline 66. Each of the offset gears 74 includes gear teeth 78 engaged with (e.g., meshed with) the main gear 72 (e.g., the gear teeth 76) or the gear teeth 78 of another of the offset gears 74. Each of the offset gears 74 is rotatable about a respective rotational axis 80. A first offset gear 74A is fixedly mounted to the shaft 48, thereby coupling the rotor 42 (see FIG. 1) to the gear assembly 64. One or more second offset gears 74B are coupled to respective accessory load assemblies 24 to drive the respective accessory load assemblies 24 in response to rotation of the main gear 72. Each of the offset gears 74 may be sized (e.g., relative to the main gear 72) to facilitate a suitable rotation speed (e.g., speed ratio) for the different rotational components (e.g., the propulsor 34, the electric motor 38, the accessory load assemblies 24, etc.) coupled to the gear assembly 64. The gear assembly 64 of FIGS. 2 and 3 includes five offset gears 74, however, the present disclosure is not limited to any particular number, arrangement, size, or other configuration of the offset gears 74.

(19) The propulsor 34 is configured for rotation about a rotational axis. The rotational axis of the propulsor 34 may be the axial centerline 66 as shown in FIG. 2, or another rotational axis which is different than the axial centerline 66. The propulsor 34 of FIG. 2 is configured as a propeller. For example, the propulsor 34 of FIG. 2 includes a hub 81 and a plurality of propeller blades 82. The propeller blades 82 are mounted to the hub 81. The propeller blades 82 are circumferentially distributed about the hub 81, for example, relative to the axial centerline 66. The propeller blades

82 may be configured as variable-pitch propeller blades. For example, the propeller blades **82** may each be rotatable about a lengthwise axis to control (e.g., selectively vary) a pitch (e.g., an angle; sometimes referred to as a “beta angle”) of the propeller blades **82**. The present disclosure, however, is not limited to propeller configurations for the propulsor **34** and the propulsor **34** may alternatively be configured as a fan (e.g., for a turbofan propulsion system), an open rotor propulsor, or another configuration of aircraft propulsion rotor.

(20) Each of the accessory load assemblies **24** may be disposed within or axially adjacent the gearbox housing **60**. For example, the accessory load assemblies **24** may be disposed at (e.g., on, adjacent, or proximate) an aft end of the gearbox housing **60**. Each of the accessory load assemblies **24** includes at least one accessory load **84**. The at least one accessory load **84** may be mounted to or otherwise supported by the gearbox housing **60**. For example, the at least one accessory load **84** may be mounted to an exterior or an interior of the gearbox housing **60**. The accessory load assemblies **24** may be disposed axially coincident with the electric motor **38**. Each of the accessory load assemblies **24** at least one accessory load **84**. Each of the accessory load assemblies **24** may further include a shaft **86**. The at least one accessory load **84** of each of the accessory load assemblies **24** is coupled to a respective one of the gears **72**, **74** of the gear assembly **64**, for example, by the shaft **86**. The at least one accessory load **84** may be mounted to and driven by the main gear **72**. For example, the at least one accessory load **84** may be mounted to the main gear **72** by the shaft **86**. Alternatively, the at least one accessory load **84** may be mounted to the main gear **72** by the output shaft **62** (e.g., the at least one accessory load **84** may be mounted to the second axial end **70**). The at least one accessory load **84** may be mounted to and driven by one of the offset gears **74**. For example, the at least one accessory load **84** may be mounted to one of the offset gears **74** by the shaft **86**.

(21) The at least one accessory load **84** for each of the accessory load assemblies **24** facilitates one or more support functions of the aircraft **1000** or its propulsion system **20** (see FIG. 1). FIGS. 2-3 illustrate examples of the at least one accessory load **84** including, but not limited to, a propeller control unit (PCU) **88**, an air compressor unit **90**, an electrical generation unit **92** (e.g., a low-voltage generator), a hydraulic pump **94**, and one or more oil pumps **96**. The present disclosure, however, is not limited to the aforementioned exemplary accessory loads **84** and the at least one accessory load **84** may include additional or alternative rotational loads or other loads. For configurations of the propulsion system **20** in which the propulsor **34** includes variable-pitch propeller blades **82**, the PCU **88** may control (e.g., modulate) the pitch of the propeller blades **82**. For example, the PCU **88** may control the pitch of the propeller blades **82** to achieve a desired thrust of the propulsion system **20** (see FIG. 1) while the propulsor **34** rotation speed remains constant or substantially constant (e.g., a constant-speed propeller assembly). The air compressor unit **90** may supply high-pressure air for the aircraft **1000** and/or its propulsion system **20**. For example, the high-pressure air may be supplied to an environmental control system (ECS) of the aircraft **1000**, for pneumatic systems of the aircraft **1000** and/or the propulsion system **20**, for air cooling systems, etc. The electrical generation unit **92** may be used to generate and supply electrical power (e.g., low-voltage electrical power), for example, to one or more electronic control systems of the aircraft **1000** and/or the propulsion system **20**. The hydraulic pump **94** may be used, for example, to supply a pressurized hydraulic fluid to the aircraft **1000** and/or the propulsion system **20** to effect operation of one or more hydraulic-actuated systems such as aircraft control surfaces, aircraft landing gear assemblies, thrust reverser assemblies, and the like. The oil pumps **96** may be configured to supply oil to components of the rotational assembly **22** such as, but not limited to, the electric motor **38**, bearing assemblies, the gear assembly **64**, and/or the PCU **88**. The oil pumps **96** may additionally or alternatively be configured as a scavenge pump to return oil from the aforementioned components of the rotational assembly **22** to an oil tank or other oil storage component. The oil pumps **96** are positioned within the gearbox housing **60**, for example, within or at (e.g., on, adjacent, or proximate) the oil cavity **61**.

(22) Referring again to FIG. 1, the electric power source **26** is electrically connected to the electric motor **38** to supply electric power to the electric motor **38** for driving the TGB module **32** (e.g., the gear assembly **64**). The electric power source **26** may include an electric generating source such as, but not limited to, an auxiliary power unit (APU) or a fuel cell generator (e.g., a hydrogen fuel cell generator). The electric power source **26** may additionally or alternatively include an electric power storage device such as, but not limited to, one or more batteries and/or one or more capacitors. The present disclosure is not limited to any particular configuration of the electric power source **26** provided the electric power source **26** can supply electric power to the electric motor **38** for operation of the electric motor. The electric power source **26** may be disposed in the propulsion system **20** or the aircraft **1000** on which the propulsion system **20** is mounted.

(23) In operation, the electric motor **38** drives rotation of the rotor **42** about the rotational axis **36**. The rotor **42**, coupled with the gear assembly **64** (e.g., the first offset gear **74A**) by the shaft **48**, drives the gear assembly **64** and, thereby, the propulsor **34** and the accessory load assemblies **24**.

(24) The present disclosure electric motor assembly **28** and TGB module **32** may facilitate conversion of a gas turbine engine propulsion system to an electric propulsion system (e.g., a propulsion system using only electrical power for propulsion). The configuration of the radially offset electric motor assembly **28** (e.g., from the axial centerline **66**) and the accessory load assemblies **24** axially coincident with the electric motor assembly **28** may accommodate limited axial space of the propulsion system **20** which may be available for components of the electric motor assembly **28** and TGB module **32**, as will be discussed in further detail. The configuration of the TGB module **32** and the accessory load assemblies **24** may additionally facilitate improved reliability of components of the propulsion system **20**. For example, in the event of a failure of the electrical power source **26**, the electric motor assembly **28**, or another component which prevents or otherwise inhibits operation of the electric motor **38**, the propulsor **34** may be operated in a feathering mode, such that rotation of the propulsor **34** (e.g., driven by ambient air flow) drives the output shaft **62** and the accessory load assemblies **24**. Accordingly, accessory loads **84** such as the PCU **88**, the air compressor unit **90**, the electrical generation unit **92**, the hydraulic pump **94**, and the oil pumps **96** may still be available for operation to supply propulsion system **20** and/or aircraft **1000** loads.

(25) Referring to FIGS. 4 and 5, in some embodiments, the electric motor assembly **28** may include a mount assembly **98** configured to support the electric motor **38** in its radially offset position relative to the TGB module **32** and the axial centerline **66**. FIG. 4 illustrates a side, cutaway view of the electric motor assembly **28** and the TGB module **32**. FIG. 5 illustrates a front, cutaway view of the electric motor assembly **28** and the TGB module **32**. The mount assembly **98** of FIGS. 4 and 5 includes a motor bracket **100** and a plurality of struts **102**. The motor bracket **100** is fixedly mounted to the motor housing **40**, for example, at an aft end of the motor housing **40**. The plurality of struts **102** of FIGS. 4 and 5 includes an upper strut **104**, a first lateral strut **106**, and a second lateral strut **108**. Each of the upper strut **104**, the first lateral strut **106**, and the second lateral strut **108** extend between and to a first axial end **110** of the struts **104**, **106**, **108** and a second axial end **112** of the struts **104**, **106**, **108**. The struts **104**, **106**, **108** are fixedly mounted to the gearbox housing **60** at their respective first axial ends **110**. For example, the first axial end **110** of the upper strut **104** may be mounted to a vertically upper portion (e.g., a mounting pad) of the gearbox housing **60** (e.g., with respect to a normal attitude of the propulsion system **20**; see FIG. 1). The first axial end **110** of the first lateral strut **106** may be mounted to a first lateral portion (e.g., an outer radial portion) of the gearbox housing **60**. The first axial end **110** of the second lateral strut **108** may be mounted to a second lateral portion (e.g., an outer radial portion) of the gearbox housing **60** opposite the first lateral portion. The second axial end **112** of each of the struts **104**, **106**, **108** may be mounted to or otherwise disposed at (e.g., on, adjacent, or proximate) the motor bracket **100**. The configuration of the mount assembly **98** may facilitate a reduction in bending moment of the electric motor assembly **28** (e.g., the motor housing **40**, the rotor **42**, and/or the shaft

48) at the interface between the electric motor assembly **28** and the gearbox housing **60**.

(26) Referring to FIG. **6**, in some embodiments, the PCU **88** may be coupled to an axially intermediate portion of the output shaft **62**, for example, forward of the gear assembly **64**. FIG. **6** illustrates a side, cutaway view of the accessory load assemblies **24** and the TGB module **32**. The output shaft **62** of FIG. **6** forms or otherwise includes a bevel gear **114** on a radial exterior of the output shaft **62**. The bevel gear **114** extends circumferentially about (e.g., completely around) the axial centerline **66**. The bevel gear **114** may be disposed between (e.g., axially between) and spaced from the first axial end **68** and the second axial end **70**, for example, within the gearbox housing **60**. The bevel gear includes gear teeth **116**. The PCU **88** of FIG. **6** is disposed outside the gearbox housing **60**. The PCU **88** of FIG. **6** is mounted to or otherwise disposed at (e.g., on, adjacent, or proximate) the gearbox housing **60**, for example, on a forward end of the gearbox housing **60**. The PCU **88** is coupled to the bevel gear **114** by a tower shaft **118** and a pinion gear **120**. The tower shaft **118** interconnects the PCU **88** and the pinion gear **120**. The pinion gear **120** includes gear teeth **122**. The gear teeth **122** are engaged with (e.g., meshed with) the bevel gear **114** (e.g., the gear teeth **116**). The PCU **88** of FIG. **6** may be supplied with pressurized oil from a sleeve extending radially through the output shaft **62**.

(27) Referring to FIG. **7**, in some embodiments, the rotational equipment assembly **22** may include more than one electric motor assembly **28**. For example, the rotational equipment assembly **22** of FIG. **7** includes two electric motor assemblies **28** (e.g., a first electric motor assembly **28A** and a second electric motor assembly **28B**). The rotor **42** of each of the first electric motor **28A** and the second electric motor **28B** is mounted to a respective one of the offset gears **74** (e.g., by the shaft **48**). The rotational equipment assembly **22** of FIG. **7**, having more than one electric motor assembly **28** coupled to the gear assembly **64**, may facilitate improved reliability and/or thrust output of the propulsion system **20** (see FIG. **1**).

(28) Referring to FIG. **8**, in some embodiments, the gear assembly **64** may further include a ring gear **124**. The ring gear **124** extends circumferentially about (e.g., completely around) the axial centerline **66**. The ring gear **124** circumscribes the offset gears **74**. The ring gear **124** includes gear teeth **126** engaged with (e.g., meshed with) the gear teeth **78** for the offset gears **74**. The ring gear **124** is fixedly mounted to the output shaft **62**. Rotation of the offset gears **74** drives rotation of the ring gear **124** about the axial centerline **66**, thereby driving rotation of the output shaft **62**. In the embodiment of FIG. **8**, the output shaft **62** may extend through the main gear **72** but may not be rotationally fixed relative to the main gear **72**.

(29) FIG. **9** illustrates an exemplary propulsion system **900** for an aircraft, such as the aircraft **1000**, which exemplary propulsion system **900** may be modified in accordance with aspects of the present disclosure. The propulsion system **900** of FIG. **9** includes a gas turbine engine **902**. The gas turbine engine **902** of FIG. **9** includes a first rotational assembly **904** and a second rotational assembly **906** arranged sequentially along an axial centerline **908** of the gas turbine engine **902**. The first rotational assembly **904** (e.g., a high-pressure spool) includes a first shaft **910**, a bladed compressor rotor **912**, and a bladed first turbine rotor **914**. The first shaft **910** interconnects the bladed compressor rotor **912** and the bladed first turbine rotor **914**. The first shaft **910** is coupled to an accessory gearbox (AGB) assembly **916**. Rotation of the bladed first turbine rotor **914** about the axial centerline **908** drives the AGB assembly **916** which, in turn, drives at least one accessory load **918** such as, but not limited to, accessory loads of the propulsion system **900**. The second rotational assembly **906** (e.g., a low-pressure spool or a power spool) includes a second shaft **920** and a bladed second turbine rotor **922**. The second shaft **920** is connected to the bladed second turbine rotor **922**. The second shaft **920** is coupled to a reduction gearbox (RGB) assembly **924**. Rotation of the bladed second turbine rotor **922** about the axial centerline **908** drives the RGB assembly **924** which, in turn, drives rotation of a propulsor **926** to generate thrust for the propulsion system **900**.

(30) Still referring to FIG. **9**, the propulsion system **900** may have a number of shape and/or size constraints which may further complicate conversion to an electric propulsion system configuration

(e.g., the propulsion system **20**). For example, a housing **928** (e.g., nacelle) of the propulsion system **900** may have an aerodynamic shape to facilitate optimal propulsor **926** thrust efficiency, pilot visibility, and aerodynamic drag. Accordingly, modification of a size and/or shape of the housing **928** to accommodate an electric propulsion system configuration may be undesirable. Moreover, modification or replacement of the housing **928** to accommodate an electric propulsion system configuration may be expensive and time consuming. The aircraft **1000** may include a cabin firewall **930** between the propulsion system **900** and occupied areas of the aircraft **1000**, which cabin firewall **930** may limit an axial extent of an electric propulsion system configuration. The location of the propulsor **926** (e.g., a propeller plane axial position **932**) may further limit the axial extent of an electric propulsion system configuration. In accordance with the present disclosure, the electric motor assembly **28** and the TGB module **32** may be used to replace the gas turbine engine **902** for propulsion and accessory loads of the propulsion system **20**, **900** while fitting within the narrow allowable space which may be available for at least some propulsion system configurations.

(31) While the principles of the disclosure have been described above in connection with specific apparatuses and methods, it is to be clearly understood that this description is made only by way of example and not as limitation on the scope of the disclosure. Specific details are given in the above description to provide a thorough understanding of the embodiments. However, it is understood that the embodiments may be practiced without these specific details.

(32) It is noted that the embodiments may be described as a process which is depicted as a flowchart, a flow diagram, a block diagram, etc. Although any one of these structures may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc.

(33) The singular forms “a,” “an,” and “the” refer to one or more than one, unless the context clearly dictates otherwise. For example, the term “comprising a specimen” includes single or plural specimens and is considered equivalent to the phrase “comprising at least one specimen.” The term “or” refers to a single element of stated alternative elements or a combination of two or more elements unless the context clearly indicates otherwise. As used herein, “comprises” means “includes.” Thus, “comprising A or B,” means “including A or B, or A and B,” without excluding additional elements.

(34) It is noted that various connections are set forth between elements in the present description and drawings (the contents of which are included in this disclosure by way of reference). It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. Any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option.

(35) No element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprise”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(36) While various inventive aspects, concepts and features of the disclosures may be described and illustrated herein as embodied in combination in the exemplary embodiments, these various aspects, concepts, and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the present application. Still further, while various alternative embodiments as to the various aspects, concepts,

and features of the disclosures—such as alternative materials, structures, configurations, methods, devices, and components, and so on—may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed. Those skilled in the art may readily adopt one or more of the inventive aspects, concepts, or features into additional embodiments and uses within the scope of the present application even if such embodiments are not expressly disclosed herein. For example, in the exemplary embodiments described above within the Detailed Description portion of the present specification, elements may be described as individual units and shown as independent of one another to facilitate the description. In alternative embodiments, such elements may be configured as combined elements.

Claims

1. An assembly for a propulsion system of an aircraft, the assembly comprising: a gearbox module including a gear assembly and an output shaft, the output shaft includes a first axial end and a second axial end, the gear assembly includes a main gear and a plurality of offset gears, the plurality of offset gears are directly engaged with the main gear, the output shaft and the main gear are configured for rotation about an axial centerline of the gear assembly; an electric motor assembly including an electric motor, the electric motor includes a rotor, the rotor is mounted to a first offset gear of the plurality of offset gears, the rotor is configured for rotation about a rotational axis to drive rotation of the gear assembly and the output shaft; a first accessory load assembly including at least one first accessory load, the at least one first accessory load is mounted to a second offset gear of the plurality of offset gears; a propulsor coupled to the first axial end of the output shaft; and a second accessory load assembly including at least one second accessory load, the at least one second accessory load is mounted to the second axial end of the output shaft.
2. The assembly of claim 1, wherein the rotational axis is radially offset from the axial centerline.
3. The assembly of claim 1, wherein the output shaft is fixedly mounted to the main gear.
4. The assembly of claim 1, wherein the first accessory load assembly is axially coincident with the electric motor assembly.
5. The assembly of claim 1, wherein the second accessory load assembly is axially coincident with the electric motor assembly.
6. The assembly of claim 1, wherein the output shaft includes a bevel gear, the assembly further comprising a third accessory load assembly including at least one third accessory load, the at least one third accessory load is engaged with the bevel gear.
7. The assembly of claim 1, wherein the gear assembly further includes a ring gear engaged with each of the offset gears, the ring gear is mounted to the output shaft and configured to drive the output shaft for rotation about the axial centerline.
8. The assembly of claim 1, wherein: the gearbox module includes a gearbox housing surrounding the gear assembly and the output shaft; the electric motor assembly includes a motor housing; and the motor housing is mounted to the gearbox housing.
9. The assembly of claim 8, wherein the electric motor assembly further includes a mount assembly mounting the motor housing to the gearbox housing, the mount assembly includes a motor bracket and a plurality of struts, the motor bracket is mounted to the motor housing at an aft end of the motor housing, each of the plurality of struts extends between and connects the motor bracket to the gearbox housing.
10. The assembly of claim 1, wherein the first accessory load assembly includes an oil pump disposed at a vertical bottom of the gearbox module.
11. The assembly of claim 10, wherein the gearbox module includes a gearbox housing surrounding the gear assembly and the output shaft, and the oil pump is disposed within the gearbox housing.

12. An assembly for a propulsion system of an aircraft, the assembly comprising: a propulsor; a gearbox module including a gear assembly and an output shaft, the gear assembly includes a main gear and a plurality of offset gears, the main gear is configured for rotation about an axial centerline of the gear assembly, the plurality of offset gears are directly engaged with the main gear, the plurality of offset gears includes a first offset gear and a plurality of second offset gears, the output shaft extends along the axial centerline in a first direction and interconnects the propulsor and the main gear; an electric motor assembly including an electric motor, the electric motor includes a rotor, the rotor is mounted to the first offset gear, the rotor is configured for rotation about a rotational axis to drive rotation of the first offset gear; a plurality of accessory load assemblies, each accessory load assembly of the plurality of accessory load assemblies including at least one accessory load coupled to a respective second offset gear of the plurality of second offset gears; and a second accessory load assembly including at least one second accessory load, the output shaft extends along the axial centerline in a second direction, opposing the first direction, and interconnects the at least one second accessory load and the main gear.

13. The assembly of claim 12, wherein the plurality of accessory load assemblies are axially coincident with the electric motor assembly.

14. The assembly of claim 12, wherein each offset gear of the plurality of offset gears is disposed radially adjacent the main gear with respect to the axial centerline.

15. The assembly of claim 12, wherein the rotational axis is radially offset from the axial centerline.

16. An assembly for a propulsion system of an aircraft, the assembly comprising: a propulsor; a gearbox module including a gear assembly and an output shaft, the gear assembly includes a main gear and a plurality of offset gears, the plurality of offset gears are directly engaged with the main gear, the plurality of offset gears includes a first offset gear and a plurality of second offset gears, the output shaft interconnects the propulsor and the main gear; an electric motor assembly including an electric motor, the electric motor includes a rotor, the rotor is mounted to the first offset gear, the rotor is configured for rotation about a rotational axis to drive rotation of the first offset gear; and a plurality of accessory load assemblies, each accessory load assembly of the plurality of accessory load assemblies including at least one accessory load coupled to a respective gear of the main gear or the plurality of offset gears, a first accessory load assembly of the plurality of load assemblies including a first accessory load, the first accessory load mounted to the output shaft.

17. The assembly of claim 16, wherein the gearbox module includes a gearbox housing surrounding the gear assembly and the output shaft.

18. The assembly of claim 17, wherein the plurality of accessory load assemblies includes a second accessory load assembly including a second accessory load and a third accessory load, the second accessory load disposed within the gearbox housing and the third accessory load disposed outside the gearbox housing.

19. The assembly of claim 18, wherein the second accessory load is an oil pump disposed at a vertical bottom of the gearbox housing.
